

Protocol Comparison Brief

Introduction

Polyhouse telemetry systems continuously monitor environmental parameters such as temperature, humidity, soil moisture, and air quality. Communication protocols play a major role in transmitting sensor data efficiently between edge devices and cloud servers. This document compares HTTP, MQTT, and CoAP.

Comparison Table

Feature	HTTP	MQTT	CoAP
Communication Model	Request-Response	Publish-Subscribe	Request-Response
Latency	Medium	Low	Very Low
Protocol Overhead	High	Low	Very Low
Bandwidth Usage	High	Low	Very Low
Offline Behavior	Poor	Good (QoS & retained messages)	Limited
Security	TLS/HTTPS	TLS	DTLS
Power Consumption	High	Low	Very Low
Reliability	Moderate	High with QoS levels	Moderate
Best Use Case	Web applications	IoT telemetry	Constrained devices

Analysis

HTTP

HTTP is widely used and easy to implement. However, it generates significant overhead because every communication requires a complete request-response cycle. This increases bandwidth usage and power consumption.

MQTT

MQTT is specifically designed for IoT applications. It follows a publish-subscribe architecture through a broker. MQTT provides low latency, low bandwidth consumption, and reliable message delivery using Quality of Service (QoS) levels.

CoAP

CoAP is a lightweight protocol designed for constrained devices and low-power networks. It uses UDP instead of TCP, reducing overhead and latency. However, it offers fewer reliability features compared to MQTT.

Recommendation for Zelbytes Edge Nodes

For Zelbytes edge nodes deployed in polyhouse telemetry systems, **MQTT is the recommended protocol** because:

- Low communication overhead
- Low latency for real-time monitoring
- Efficient bandwidth utilization
- Reliable message delivery through QoS
- Better handling of intermittent network connectivity
- Widely adopted in IoT ecosystems

Conclusion

While HTTP is suitable for traditional web communication and CoAP is ideal for highly constrained devices, MQTT provides the best balance of reliability, efficiency, and scalability for polyhouse telemetry applications. Therefore, MQTT is the preferred choice for Zelbytes edge nodes.